Realised Volatility Forecasting — Multi-Asset Benchmark

Direct Multi-Step Models: HAR, NLinear, DLinear, SAS, RNN

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1 Setup

$x_t = \log \mathrm{RV}_t$, z-scored per rolling window (μ, σ frozen until refit). Direct multi-step: one model per $h \in \{1, 5, 10\}$. Ridge $\alpha \in \{10^k : k \in [-4, 5],\ \Delta k{=}0.25\}$ (37 values), 5-fold CV.

2 Models

2.1 HAR — Heterogeneous Autoregressive Model (Corsi 2009)

Features. Let $\bar{x}_t^{(w)} = \frac{1}{5}\sum_{k=0}^{4} x_{t-k}$ and $\bar{x}_t^{(m)} = \frac{1}{22}\sum_{k=0}^{21} x_{t-k}$.

$$\hat{x}_{t+h} = \alpha + \beta_d\, x_t + \beta_w\, \bar{x}_t^{(w)} + \beta_m\, \bar{x}_t^{(m)} \tag{1}$$

HAR-OLS: $\min_\beta \|X\beta - y\|_2^2$. **HAR-Ridge**: column-normalise $[x^{(d)}, x^{(w)}, x^{(m)}]$; penalise slopes only; back-transform $w = w_{\text{scaled}}/\sigma_{\text{col}}$.

2.2 NLinear and DLinear (Zeng et al. 2023)

Let $\boldsymbol{x}_t = (x_{t-L+1}, \ldots, x_t)^\top \in \mathbb{R}^L$ be the lookback window.

NLinear.

$$\tilde{\boldsymbol{x}}_t = \boldsymbol{x}_t - x_t, \qquad \hat{x}_{t+h} = \tilde{\boldsymbol{x}}_t^\top w_h + x_t \tag{2}$$

DLinear.

$$\boldsymbol{\tau}_t = \mathrm{MA}_k(\boldsymbol{x}_t), \quad \boldsymbol{\varsigma}_t = \boldsymbol{x}_t - \boldsymbol{\tau}_t, \qquad \hat{x}_{t+h} = \begin{bmatrix} \boldsymbol{\tau}_t \\ \boldsymbol{\varsigma}_t \end{bmatrix}^\top w_h \tag{3}$$

2.3 GARCH — AR(p)-GARCH(1,1)

$x_t = \mu + \sum_{i=1}^{p} \phi_i x_{t-i} + \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_G \varepsilon_{t-1}^2 + \beta_G \sigma_{t-1}^2$. Multi-step: iterate variance recursion analytically. QMLE.

2.4 SAS — State Affine System

Diagonal basis. $P_0, \ldots, P_p \in \mathbb{R}^n,\ Q_0, \ldots, Q_q \in \mathbb{R}^n,\ z \in \mathbb{R}$.

$$P(z) = \mathrm{clip}\Big(\sum_{d=0}^{p} P_d\, z^d,\ -1{+}\epsilon,\ 1{-}\epsilon\Big) \in (-1,1)^n \tag{4}$$

$$Q(z) = \sum_{d=0}^{q} Q_d\, z^d \in \mathbb{R}^n \tag{5}$$

$$s_t = P(z_t) \odot s_{t-1} + Q(z_t) \tag{6}$$

Clipping ensures $|P(z)_i| < 1$ (unconditional stability). Readout: $\hat{x}_{t+h} = s_t^\top w_h$, W_h fitted by ridge regression with 5-fold CV.

Transition polynomial P. Let ρ be the spectral-norm bound and $\delta = \max(0.005,\ (1{-}\rho)\cdot 0.3)$ a small headroom margin.

- $P_0[i] \overset{\text{iid}}{\sim} \mathcal{U}(-\rho,\ \rho)$ — base eigenvalues spread uniformly to encode timescales from fast-decaying to near-unit-root.
- $P_k[i] \overset{\text{iid}}{\sim} \mathcal{N}\big(0,\ \delta/5^k\big)$ for $k \geq 1$ — exponentially shrinking input-modulation terms so that the clip boundary is rarely reached for $|z| \leq 5$.

Input-drive polynomial Q. The injection $Q(z_t)$ adds noise with variance $\propto \sum_k Q_k[i]^2\, \mathbb{E}[z^{2k}]$. For $z \sim \mathcal{N}(0,1)$, $\mathbb{E}[z^{2k}] = (2k-1)!! = 1\cdot 3\cdot 5\cdots(2k{-}1)$ (double factorial). To keep injection variance roughly uniform across degrees:

$$Q_k[i] \overset{\text{iid}}{\sim} \mathcal{N}\left(0,\ \frac{1}{\sqrt{n\,(2k-1)!!}}\right) \tag{7}$$

so $k{=}0,1$ use scale $1/\sqrt{n}$ (unchanged from a linear SAS), while $k{=}2$ is shrunk by $1/\sqrt{3} \approx 0.58$ and $k{=}3$ by $1/\sqrt{15} \approx 0.26$. This prevents state explosion for large inputs without hand-tuning per-degree scales.

Associative structure. $\ell_t = (P(z_t), Q(z_t))$. Composition rule:

$$(A_j,\ b_j) \circ (A_i,\ b_i) \ =\ \big(A_j \odot A_i,\ \ A_j \odot b_i + b_j\big) \tag{8}$$

is associative $\Rightarrow$ parallel divide-and-conquer scan.

Algorithm 1: Two-Level Parallel Scan — compute $s_1, \ldots, s_T$ from $z_1, \ldots, z_T$

Input : Transitions $\ell_1, \ldots, \ell_T$;;
chunk size B;;
$s_0 = \mathbf{0}$
Output : States $s_1, \ldots, s_T$;;
s_T

$K \leftarrow \lceil T/B \rceil$; pad to KB
Phase 1 (vmap over K chunks);
  $\forall\, k:\ (A^{\text{intra}}_{k,\cdot},\ b^{\text{intra}}_{k,\cdot}) \leftarrow \textsc{AssocScan}(\circ,\ \ell_{kB+1}, \ldots, \ell_{(k+1)B})$
Phase 2 (inter-chunk scan over K endpoints);
  $(A^{\text{inter}},\ b^{\text{inter}}) \leftarrow \textsc{AssocScan}(\circ,\ (A^{\text{intra}}_{1,B}, b^{\text{intra}}_{1,B}), \ldots)$
Phase 3 (carry for chunk $k \geq 1$);
  $c_0 = s_0$;;
$c_k = A^{\text{inter}}_{k-1} \odot s_0 + b^{\text{inter}}_{k-1}$
Phase 4 (vmap over K chunks);
  $\forall\, k, j:\ s_{kB+j} = A^{\text{intra}}_{k,j} \odot c_k + b^{\text{intra}}_{k,j}$

Depth $O(\log T)$ vs. $O(T)$ sequential.

2.5 RNN — LSTM / GRU / LRU

Readout: $\hat{x}_{t+h} = \text{Linear}_h(h_T)$, one head per h. Training: Adam, early stopping (patience=20, val=20% chronological).

LRU (Orvieto et al. 2023):

$$h_t = \sigma(\nu) \odot h_{t-1} + Bz_t, \qquad y_t = Ch_t, \qquad \sigma(\nu) \in (0,1)^d \tag{9}$$

3 Evaluation Protocol

Algorithm 2: Rolling Out-of-Sample Evaluation

Input : Series $x_1, \ldots, x_T$; window W; refit frequency R; horizons $\mathcal{H}$
Output : Squared-error sequences $\{e_{t,m}^{(h)}\}$ for each model m, horizon h

```
(μ, σ) ← (0, 1);
for t = W to T − max(H) do
    train ← x_{t−W:t};
    if t mod R = 0 then
        (μ, σ) ← FitScaler(train);
        traiñ ← (train − μ)/σ;
        foreach model m do
            m.Fit(traiñ, H)
    z_t ← (x_t − μ)/σ;
    foreach model m, horizon h ∈ H do
        ẑ_{t+h} ← m.Predict(h);
        e^{(h)}_{t,m} ← (ẑ_{t+h} − z_{t+h})^2;
    foreach model m do
        m.Update(z_t)
```

Model Confidence Set (Hansen et al. 2011). $L \in \mathbb{R}^{T \times M}$, row-normalised by cross-model mean to prevent high-vol dominance.

1. $d_{t,m} = L_{t,m} - \bar{L}_t$
2. $T_R = \max_m \bar{d}_m / \hat{\sigma}_m$ ($\hat{\sigma}_m$: circular block bootstrap, B reps)
3. Eliminate worst model if $p < \alpha$; repeat until H_0 not rejected.
4. Surviving set = MCS_α.

MCS Frequency. Run MCS per symbol per horizon; report $c_m^{(h)}/N$.

4 Hyperparameters

Table 1: Fixed hyperparameters used in all experiments.

Component	**Parameter**	**Value**
Evaluation	Training window W	2 000
	Refit frequency R	20 steps ($\approx$1 month)
	Horizons $\mathcal{H}$	$\{1, 5, 10\}$ days
MCS	Significance α	0.10
	Bootstrap replications B	1 000
HAR	Lags (d, w, m)	$(1, 5, 22)$ days
	Ridge CV folds	5
NLinear / DLinear	Lookback L	20
	DLinear MA kernel	5
GARCH	AR lags p	5
	Estimation	QMLE
SAS (diagonal)	Reservoir size n	200
	Spectral norm ρ	0.95
	P-polynomial degree p	1
	Q-polynomial degree q	1 ($\mathrm{SAS_{q1}}$) or 2 ($\mathrm{SAS_{q2}}$)
	Washout	50 steps
RNN (GRU)	Hidden size	32
	Layers	1
	Lookback	20
	Epochs (max)	500
	Early stopping	patience = 20, val = 20%
MdSAS	Groups (D+1)	2 (degrees 0,1) or 3 (degrees 0,1,2)
	n per group	50 or 100
	Spectral norm ρ	$\{0.90, 0.95\}$
	α grid (grouped)	7-value subset (Δk=1.5, 49 combos)
Ridge grid	Range	10^k, $k \in [-4, 5]$
	Spacing (single-α)	$\Delta k = 0.25$, 37 values
	Spacing (grouped)	$\Delta k = 1.5$, 7 values (49 combos/2 groups)

5 Results

5.1 All Models — 30 Symbols, $h \in \{1, 5, 10\}$

Table 2: Out-of-sample MSE — mean ± std across 30 symbols (W=2000, R=20, target: $\log \mathrm{RV}_5$, N_{sym}=30). Sorted by avg MSE. Bold: lowest mean per horizon. $\mathrm{SAS}_{\mathrm{q1/q2}}$: diagonal reservoir, n=200, ρ=0.95, p=1.

Model	$h = 1$	$h = 5$	$h = 10$	avg
$\mathrm{SAS}_{\mathrm{q2}}$	$\mathbf{0.3958 \pm 0.0718}$	$\mathbf{0.5070 \pm 0.0934}$	0.5837 ± 0.1145	**0.4955**
HAR	0.4002 ± 0.0721	0.5090 ± 0.0928	$\mathbf{0.5827 \pm 0.1053}$	0.4973
$\mathrm{SAS}_{\mathrm{q1}}$	0.3973 ± 0.0713	0.5065 ± 0.0913	0.5914 ± 0.1058	0.4984
DLinear	0.4001 ± 0.0720	0.5116 ± 0.0927	0.5855 ± 0.1057	0.4991
NLinear	0.4032 ± 0.0741	0.5191 ± 0.0973	0.5977 ± 0.1130	0.5067
GARCH	0.4104 ± 0.0741	0.5459 ± 0.0976	0.6466 ± 0.1182	0.5343

Table 3: Model Confidence Set (MCS, α=0.10, B=1000 bootstrap replications, Hansen et al. 2011) — frequency of inclusion in the superior set across 30 symbols, sorted by total. A model survives if it cannot be shown to be significantly worse than the best model on that symbol.

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
HAR	22/30 (73%)	29/30 (97%)	30/30 (100%)	81/90	0.4973
$\mathrm{SAS}_{\mathrm{q1}}$	28/30 (93%)	27/30 (90%)	23/30 (77%)	78/90	0.4984
DLinear	21/30 (70%)	18/30 (60%)	19/30 (63%)	58/90	0.4991
$\mathrm{SAS}_{\mathrm{q2}}$	15/30 (50%)	17/30 (57%)	9/30 (30%)	41/90	0.4955
NLinear	2/30 (7%)	0/30 (0%)	0/30 (0%)	2/90	0.5067
GARCH	1/30 (3%)	0/30 (0%)	0/30 (0%)	1/90	0.5343

5.2 SAS Ablations — MCS Frequency (α=0.10, $N_{\mathbf{stable}}$ symbols)

Model naming: `q`k = Q-polynomial degree k; `n`N = reservoir size N; `sn`XX = spectral norm $0.XX$; `p`k = P-polynomial degree k (default p=1 when omitted).

Table 4: Q1 — Q-degree sweep (n=200, ρ=0.95, p=1). Stable subset: 29/30 symbols (q3 diverged on .IBEX). Sorted by MCS total.

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
`q1_n200_sn95`	26/29 (90%)	26/29 (90%)	29/29 (100%)	81/87	0.4986
`q2_n200_sn95`	29/29 (100%)	27/29 (93%)	25/29 (86%)	81/87	0.4959
`q3_n200_sn95`	18/29 (62%)	18/29 (62%)	18/29 (62%)	54/87	0.5054
HAR	4/29 (14%)	5/29 (17%)	8/29 (28%)	17/87	0.4978

Table 5: Q2 — reservoir size sweep for q=2 (ρ=0.95, p=1). Stable subset: 29/30 symbols (`q2_n100` diverged on .IBEX). Sorted by MCS total.

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
`q2_n200_sn95`	25/29 (86%)	29/29 (100%)	28/29 (97%)	82/87	0.4959
`q2_n100_sn95`	28/29 (97%)	24/29 (83%)	24/29 (83%)	76/87	0.5000
`q2_n400_sn95`	26/29 (90%)	27/29 (93%)	21/29 (72%)	74/87	0.4966
`q1_n200_sn95`	10/29 (34%)	10/29 (34%)	13/29 (45%)	33/87	0.4986
HAR	0/29 (0%)	1/29 (3%)	1/29 (3%)	2/87	0.4978

Table 6: Q3 — spectral-norm sweep for q=2 (n=200, p=1). All 30 symbols stable. Sorted by MCS total.

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
`q2_n200_sn90`	30/30 (100%)	30/30 (100%)	27/30 (90%)	87/90	0.5014
`q2_n200_sn95`	24/30 (80%)	24/30 (80%)	24/30 (80%)	72/90	0.4955
`q2_n200_sn85`	18/30 (60%)	9/30 (30%)	13/30 (43%)	40/90	0.5075
`q1_n200_sn95`	13/30 (43%)	19/30 (63%)	23/30 (77%)	55/90	0.4984
HAR	1/30 (3%)	4/30 (13%)	14/30 (47%)	19/90	0.4973

5.3 SAS as Data-Generating Process — Specification Experiment

DGP construction. For each real-data symbol, we fit SAS_{q2} on the *full* real series (all T observations) to obtain initialised reservoir matrices (P, Q), readout W_1, terminal state s_T, and residual noise scale $\hat{\sigma} = \text{std}(z_{t+1} - s_t^\top W_1)_{t=\tau}^{T-1}$. We then generate N_{gen}=4,000 synthetic steps in closed-loop autoregressive form:

Algorithm 3: SAS-DGP autoregressive generation

Input : $(P, Q, W_1, s_T, \hat{\sigma})$;;
ν (Student-t df);;
clip Z_c=6

$s \leftarrow s_T$;
for $t = 1$ **to** N_{gen} **do**
  $\varepsilon \sim t(\nu)$;
  $z_t \leftarrow \text{clip}\big(s^\top W_1 + \hat{\sigma}\,\varepsilon,\ -Z_c,\ Z_c\big)$;
  $s \leftarrow P(z_t) \odot s + Q(z_t)$;

The clip prevents $Q(z)$=Q_0+$Q_1 z$+$Q_2 z^2$ from exploding under fat-tailed draws ($|z|$≤6 covers all realistic log-RV z-scores). The same rolling OOS protocol is then applied on the synthetic series (W=2000, R=20, $\mathcal{H}$={1, 5, 10}).

`SAS_diff` uses a different random seed from the DGP (different reservoir matrices, same q=2 architecture). `SAS_q1` uses q=1 (deliberate misspecification) with its own seed. Each symbol contributes three independent runs via `DGP_SEEDS` = [42, 137, 256], giving 3×N_{sym} MCS observations for greater statistical power. Each SAS estimator receives a unique initialisation seed per run so that no two SAS instances share the same random matrices.

Table 7: Q4 — P-degree with q=2 (n=200, ρ=0.95). All 30 symbols stable. Sorted by MCS total.

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
`p2q2_n200_sn95`	28/30 (93%)	28/30 (93%)	27/30 (90%)	83/90	0.4958
`q2_n200_sn95`	25/30 (83%)	27/30 (90%)	24/30 (80%)	76/90	0.4955
`q1_n200_sn95`	19/30 (63%)	20/30 (67%)	22/30 (73%)	61/90	0.4984
HAR	0/30 (0%)	1/30 (3%)	3/30 (10%)	4/90	0.4973

Table 8: DGP specification experiment — MSE. DGP: $\mathrm{SAS}_{\mathrm{q2}}$ (p=1, n=200, ρ=0.95), noise $\hat{\sigma} \cdot t(5)$, N_{gen}=4,000. Mean ± std over 89 symbol-seed runs (3 seeds × 30 symbols, one symbol skipped as too short for a second seed). Bold: lowest mean per horizon. † high cross-run variance (CV>0.5 at h=10).

Model	$h = 1$	$h = 5$	$h = 10$	avg
HAR	0.3836 ± 0.1006	0.4906 ± 0.1132	0.5669 ± 0.1270	0.4804
DLinear	0.3841 ± 0.1007	0.4934 ± 0.1135	0.5698 ± 0.1276	0.4825
`SAS_diff`	0.3817 ± 0.1029	0.4933 ± 0.1113	0.5825 ± 0.1255	0.4858
`SAS_q1`†	0.3823 ± 0.1001	0.4972 ± 0.1285	0.6909 ± 0.7466	0.5235

Table 9: DGP specification experiment — MCS frequency. Same setup as Table 8. N=89 symbol-seed runs; losses row-normalised by cross-model mean. α=0.10, B=1000 bootstrap replications. † `SAS_q1` has CV>0.5 at h=10: high variance inflates MCS inclusion despite the worst average MSE.

Model	$h = 1$	$h = 5$	$h = 10$	total	avg MSE
`SAS_q1`†	85/89 (96%)	84/89 (94%)	73/89 (82%)	242/267	0.5235
HAR	19/89 (21%)	79/89 (89%)	87/89 (98%)	185/267	0.4804
`SAS_diff`	33/89 (37%)	61/89 (69%)	45/89 (51%)	139/267	0.4858
DLinear	11/89 (12%)	38/89 (43%)	67/89 (75%)	116/267	0.4825